

分析:

0 $S' \rightarrow S$

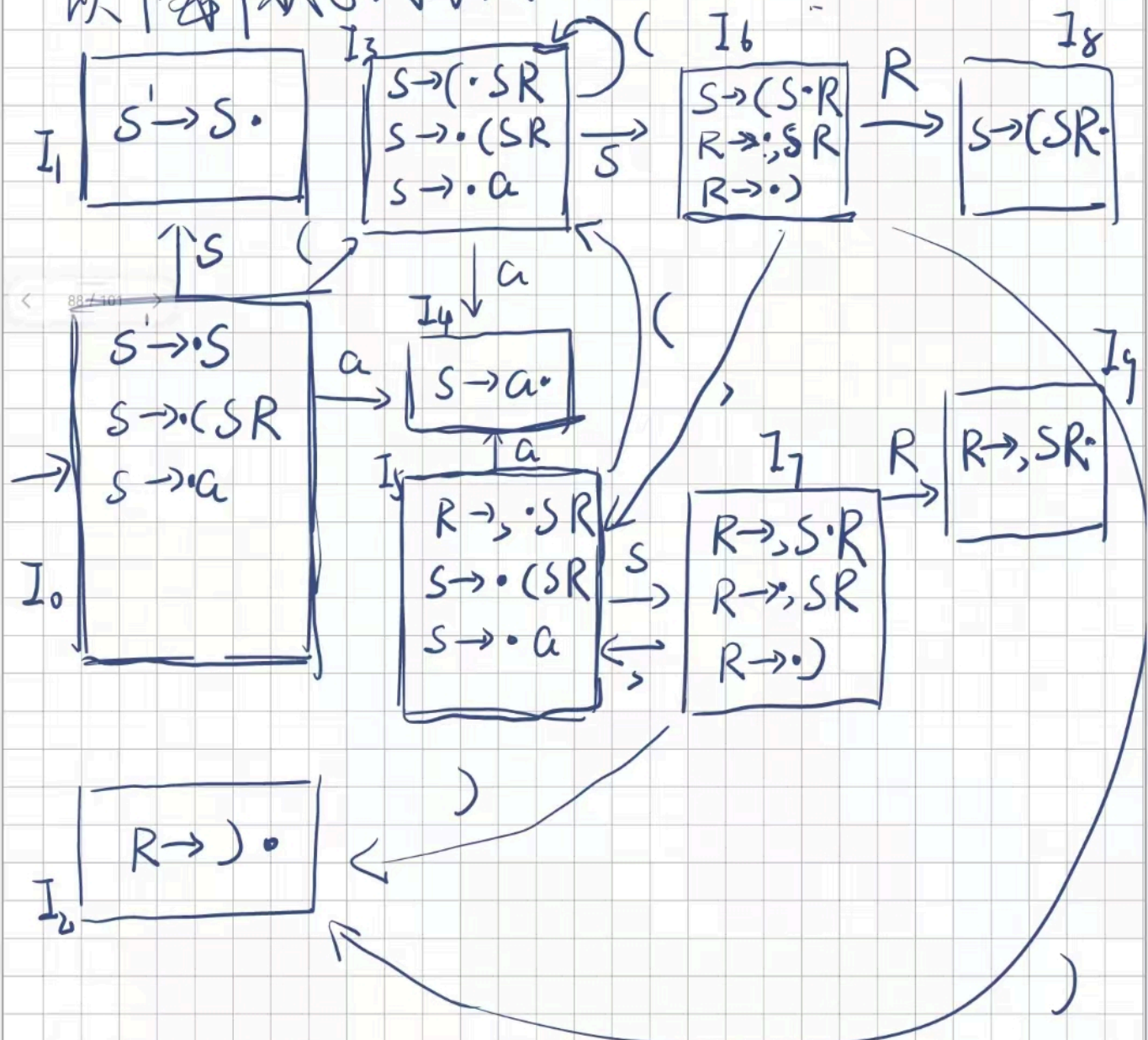
1 $S \rightarrow (SR$

2 $S \rightarrow a$

3 $R \rightarrow ,SR$

4 $R \rightarrow)$

以下每个状态均为终结状态，故省略双竖线



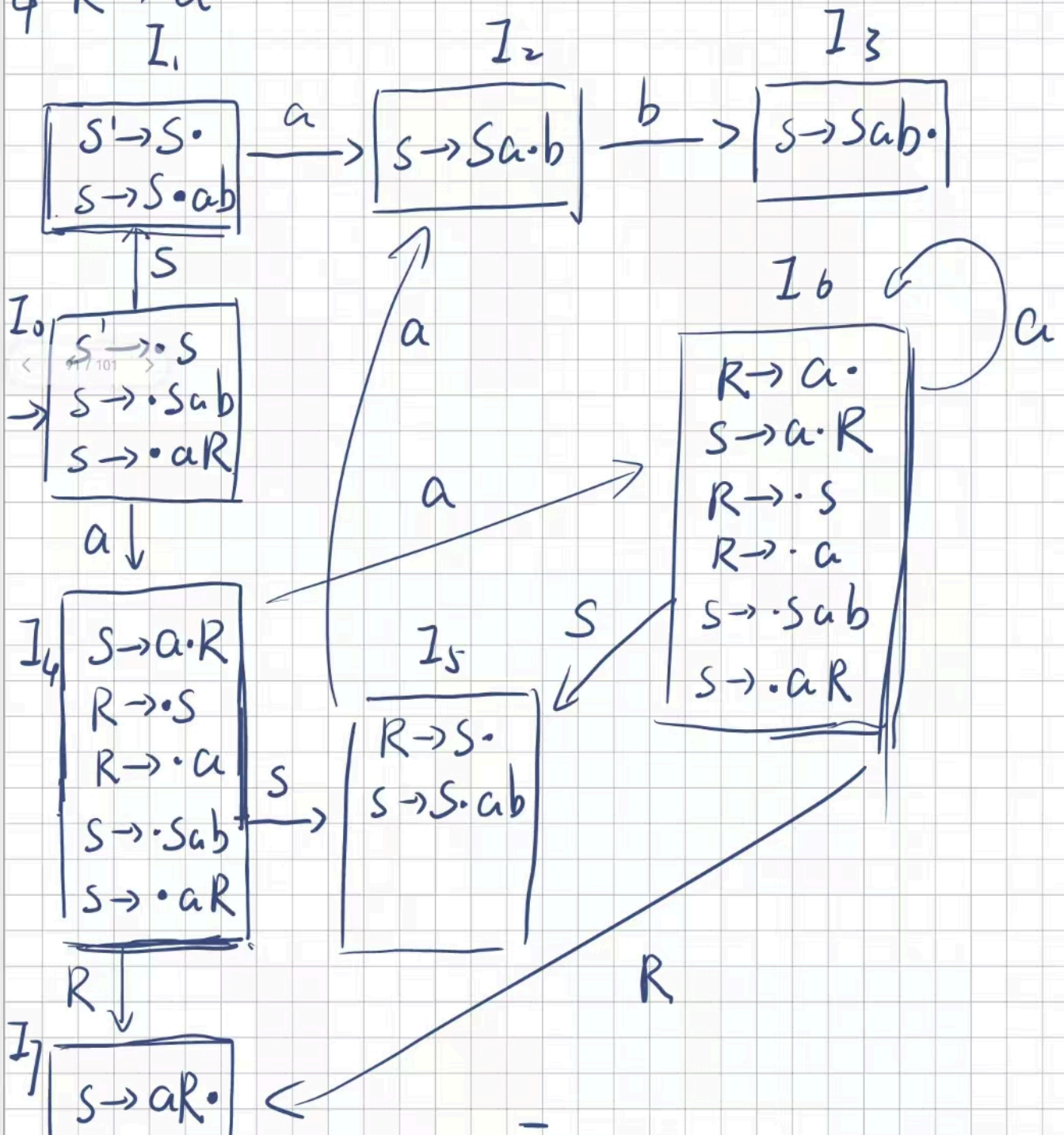
建表格如下: $FLOW(S) = \{ \$,) \}$
 $FLOW(R) = \{ \$,) \}$

state	action				goto	
	a	,	(	)	\$	R
0	S4		S3			1
1					acc	
2		r4		r4	r4	
3	S4		S3			6
4		r2		r2	r2	
5	S4		S3			7
6		S5		S2		8
7		S5		S2		9
8		r1		r1	r1	
9		r3		r3	r3	

没有冲突:

- 0 $S' \rightarrow S$
- 1 $S \rightarrow Sab$
- 2 $S \rightarrow aR$
- 3 $R \rightarrow S$
- 4 $R \rightarrow a$

以下状态均为最终状态
所以省略双线



0 $S' \rightarrow S$

1 $S \rightarrow A$

2 $A \rightarrow BA$

3 $A \rightarrow \varepsilon$

4 $B \rightarrow aB$

5 $B \rightarrow b$

I_1 $S' \rightarrow S \cdot, \$$

所有状态均为最终状态
为方便省略双线

$FIRST(A) = \{\varepsilon, a, b\}$

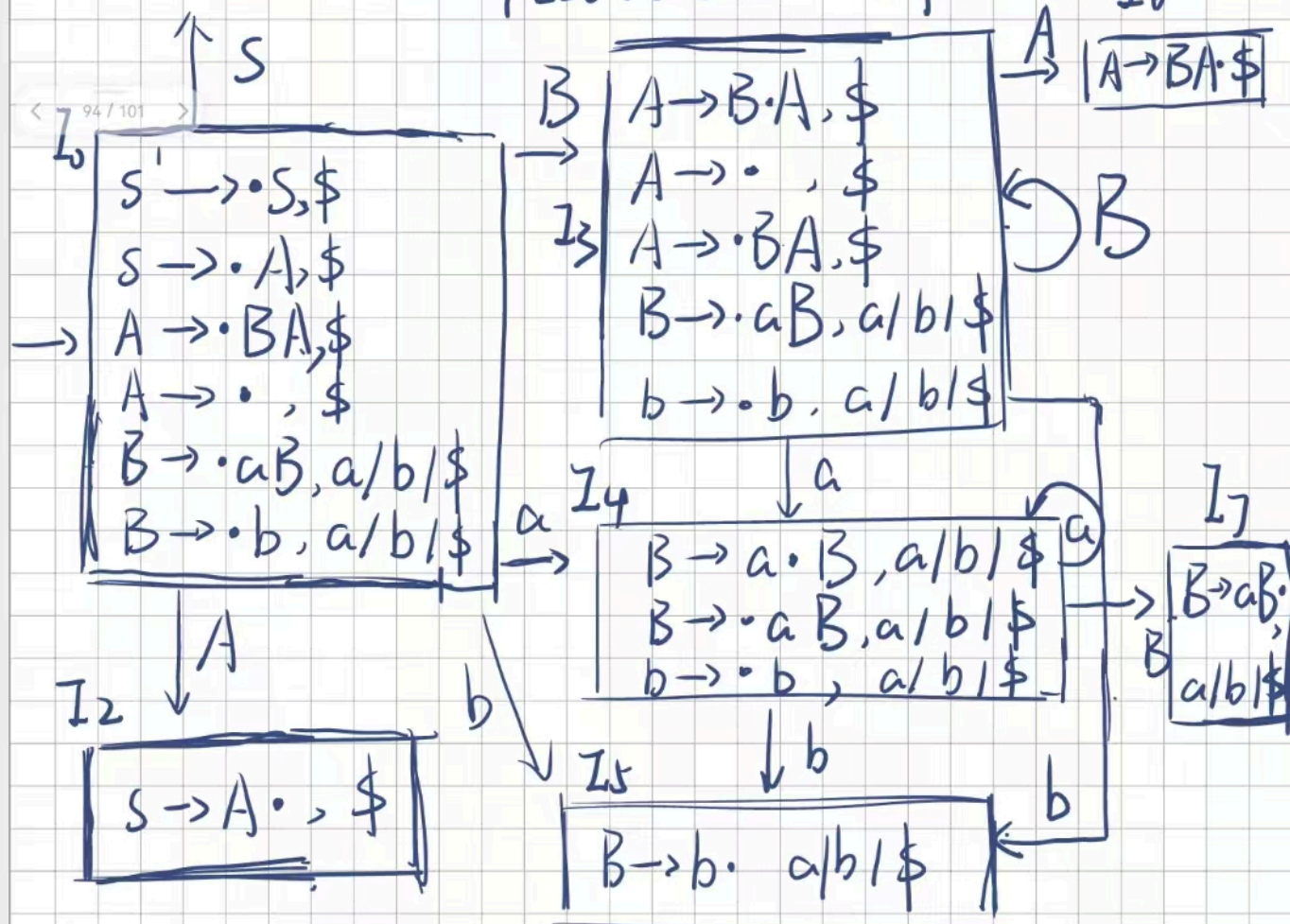
$FIRST(B) = \{a, b\}$

$FIRST(S) = \{\varepsilon, a, b\}$

$FOLLOW(A) = \{\$, \varepsilon\}$

$FOLLOW(B) = \{a, b, \$\}$

$FOLLOW(S) = \{\$, \varepsilon\}$



表格如下:

state	action			goto		
	a	b	\$	S	A	B
0	S4	S5	r3	1	2	3
1			acc			
2			r1			
3	S4	S5	r3		6	3
4	S4	S5				7
5	r5	r5	r5			
6			r2			
7	r4	r4	r4			

填表无任何冲突, 证明为 LR(1)

abab

step	symbol	state	Input	Reference	Action	Output
1.	\$	0	abab\$	$a[0, a] = s_4$	shift	
2.	\$a	04	bab\$	$a[4, b] = s_5$	shift	
3.	\$ab	045	ab\$	$a[5, a] = r_5$	reduce $B \rightarrow b$	
				$g[4, B] = 7$		
4.	\$aB	047	ab\$	$a[7, a] = r_4$	reduce $B \rightarrow aB$	
				$g[0, B] = 3$		
5.	\$B	03	ab\$	$a[3, a] = s_4$	shift	
6.	\$Ba	034	b\$	$a[4, b] = s_5$	shift	
7.	\$Bab	0345	\$	$a[5, \$] = r_5$	reduce $B \rightarrow b$	
				$g[4, B] = 7$		
8.	\$BaB	0347	\$	$a[7, \$] = r_4$	reduce $B \rightarrow aB$	
				$g[3, B] = 3$		
9.	\$BB	033	\$	$a[3, \$] = r_3$	reduce $A \rightarrow \epsilon$	
				$g[3, A] = 6$		
10.	\$BBA	0336	\$	$a[6, \$] = r_2$	reduce $A \rightarrow BA$	
				$g[3, A] = 6$		
11.	\$BA	036	\$	$a[6, \$] = r_2$	reduce $A \rightarrow BA$	
				$g[0, A] = 2$		
12.	\$A	02	\$	$a[2, \$] = r_1$	reduce $S \rightarrow A$	
13.	\$S	01	\$	$g[0, S] = 1$	accept	

$13 \downarrow a$
 $s \rightarrow Aa, \$$

不再有可合并项，构造表格如下。

state	action					goto	
	a	b	c	d	\$	S	A
0		s5		s6		1	2
1					acc		
2	s3						
3					r1		
4			s8				
5				s9			4
6	r5		s7				
7					r3		
8					r2		
9	s10		r5				
10					r4		

没有冲突，所以是 LALR(1)

对于状态 6: $c \in FOLLOW(A)$, $a[6, c] = r5$

对于 $a[6, c] = s7$ 存在 shift-reduce 冲突，不是 SLR(1)